THE QUANTITATIVE BRASCO–DE PHILIPPIS–VELICHKOV ROUTE TO SHARP QUANTITATIVE FABER–KRAHN STABILITY (SINGLE-SET SELECTION PRINCIPLE)

1. Introduction

A bounded open set $\Omega \subset \mathbb{R}^N$, $N \geq 2$, with the same volume as the ball B has a smaller torsional rigidity unless it *is* a ball; equivalently, its first Dirichlet eigenvalue $\lambda_1(\Omega)$ exceeds $\lambda_1(B)$ unless Ω is a ball. The *Saint–Venant deficit*

$$\delta_T(\Omega) \;:=\; T(B) - T(\Omega) \;\geq\; 0,$$

where $T(\cdot)$ is the torsional rigidity and B is the ball of the same volume, and the *spectral deficit* $\lambda_1(\Omega) - \lambda_1(B) \geq 0$ are the two natural quantitative versions of the corresponding rigidity statements. The *Fraenkel asymmetry*

$$\mathcal{A}_F(\Omega) \;:=\; \inf_{x_0 \in \mathbb{R}^N} \frac{\big|\Omega \triangle (x_0 + B)\big|}{|\Omega|}$$

measures, in L^1, how far Ω sits from being a translate of B. The central sharp quantitative statement,

$$\lambda_1(\Omega) - \lambda_1(B) \;\geq\; c_{\mathrm{FK}}(N,R)\, \mathcal{A}_F(\Omega)^2, \tag{1}$$

or its Saint–Venant analogue $\delta_T(\Omega) \geq c_{\mathrm{SV}}(N,R)\, \mathcal{A}_F(\Omega)^2$, is the theorem of Brasco–De Philippis–Velichkov [1]. The exponent 2 in (1) is sharp.

The non-constructive step in BDV, and what this article does. The proof of [1] proceeds by a *selection principle*: given a sequence (Ω_k) of near-counterexamples to (1) (deficit going to zero, asymmetry bounded below), one passes to a sequence of penalised quasi-minimisers $\widetilde{\Omega}_k$, exploits free-boundary regularity to show $\widetilde{\Omega}_k$ converges to a ball in a strong enough topology, and obtains a contradiction with the BDV *nearly-spherical second-variation theorem* [1, Theorem 3.3]. The sequential selection + compactness + contradiction is the only non-constructive piece in an otherwise quantitative chain.

This article assembles the four-step quantitative route that replaces that contradiction by a *single-set selection map*. Given any one set Ω_0 with small deficit and a positive smooth asymmetry $\alpha(\Omega_0)$, the selection produces a single quasi-open minimiser $\widetilde{\Omega}$ whose deficit-to-asymmetry ratio $Q_\alpha(\widetilde{\Omega}) := \delta(\widetilde{\Omega})/\alpha(\widetilde{\Omega})$ is bounded above by that of Ω_0 up to a constant; once $\widetilde{\Omega}$ is selected, the rest of the chain is constructive and gives a uniform lower bound $Q_\alpha(\widetilde{\Omega}) \geq c_*(N)$, hence (1) by computation.

The four steps. *Step 1 (Selection, §3).* A single-set quantitative selection map $\Omega_0 \mapsto \widetilde{\Omega}$ produced as a penalised minimiser of an explicit functional $J_{\tau,\hat{\eta},\Omega_0}$. The selected $\widetilde{\Omega}$ has $\alpha(\widetilde{\Omega}) \geq (1-\rho)\alpha(\Omega_0)$ and $Q_\alpha(\widetilde{\Omega}) \leq C_{\mathrm{sel}}(N,R)/(1-\rho) \cdot Q_\alpha(\Omega_0)$, with $\rho < 1$ and C_{sel} depending polynomially on the BDV penalised-minimiser regularity constants. This is the conceptual core of Plan 1 and the substitute for BDV's sequential contradiction.

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Step 2 (Graph entry, §4). Below an explicit asymmetry threshold $\alpha_{\mathrm{graph}}(N, R)$, the free boundary $\partial\widetilde{\Omega}$ is the graph of a $C^{1,\gamma}$ function over a sphere. The chain α →Hausdorff distance →flatness →BDV flatness improvement ([1, Theorem 4.18]) is fully constructive.

Step 3 (Small C^{2,γ_0} entry, §5). Uniform Schauder bounds (hodograph + Kinderlehrer–Nirenberg) give a fixed-order high $C^{m,\gamma}$ bound on the graph; interpolating that against the small L^∞/Hausdorff size of the graph yields a *small* C^{2,γ_0} norm — precisely the regularity hypothesis of the BDV nearly-spherical second-variation theorem [1, Theorem 3.3]. This is the regularity bootstrap that closes the gap between the $C^{1,\gamma}$ output of Step 2 and the input of Step 4. An alternative *Bernoulli spectral closure*, conditional on a bilinear source enumeration, is recorded as the self-contained subsection §5.1.

Step 4 (Closure + Kohler–Jobin transfer, §6). BDV Theorem 3.3 applied to a small-C^{2,γ_0} graph gives $Q_\alpha(\widetilde{\Omega}) \geq c_*(N)$. Combining with Step 1's bound on Q_α yields sharp Saint–Venant stability $\delta_T(\Omega) \geq c_{\mathrm{SV}}(N, R)\, \mathcal{A}_F(\Omega)^2$. The Kohler–Jobin transfer (§6.4) converts this to sharp Faber–Krahn stability (1), completing the proof.

Main result.

Theorem 1.1 (Sharp quantitative Faber–Krahn stability)**.** *Let $N \geq 2$ and $R > 0$. There exists a constant $c_{\mathrm{FK}}(N, R) > 0$ such that, for every bounded open $\Omega \subset \mathbb{R}^N$ with $|\Omega| = |B_1|$ and $\Omega \subset B_R$,*

$$\lambda_1(\Omega) - \lambda_1(B_1) \;\geq\; c_{\mathrm{FK}}(N, R)\, \mathcal{A}_F(\Omega)^2.$$

The constant $c_{\mathrm{FK}}(N, R)$ is given as an explicit polynomial combination of the selection constant of Theorem 3.2, the graph-entry threshold $\alpha_{\mathrm{graph}}(N, R)$, the small-$C^{2,\gamma_0}$ interpolation constant of Proposition 5.1, the BDV nearly-spherical constant $c_{\mathrm{BDV}}(N)$ of Theorem 6.1, and the Kohler–Jobin constant $c_{\mathrm{KJ}}(N, R)$ of Theorem 6.9. The exponent 2 is sharp.

Comparison with the Selection–Level–Brandolini route. The route presented here keeps BDV's selection step (now made constructive) and BDV's nearly-spherical second-variation closure, but replaces no piece of BDV with a new geometric mechanism. An alternative route, presented in a companion manuscript, replaces Steps 2–3 of the chain by a level-set deficit identity followed by the Brandolini–Nitsch–Salani–Trombetti pinching theorem and a direct radial-graph stability estimate; that route bypasses both the flatness improvement (BDV Theorem 4.18) and the nearly-spherical second-variation theorem (BDV Theorem 3.3) but introduces a more delicate radial-graph realisation step. The two routes agree on Step 1 (selection) and Step 4 (Kohler–Jobin transfer) and differ in how they reach the small-norm graph regime where the final quantitative inequality is closed.

Organisation. Section 2 collects preliminaries: notation, the BDV penalised-minimiser regularity package (used in Step 1), the BDV flatness improvement (used in Step 2), the BDV nearly-spherical second-variation theorem (used in Step 4), and the Kohler–Jobin transfer (used in the final step). Section 3 executes Step 1, opening with a flow heuristic that explains why the selected $\widetilde{\Omega}$ inherits the regularity needed downstream and then proving the quantitative single-set selection theorem. Section 4 carries out Step 2 (graph entry). Section 5 proves the small-C^{2,γ_0} Schauder bootstrap; the Bernoulli spectral closure alternative is laid out in subsection §5.1. Section 6 closes the chain with BDV's nearly-spherical theorem, the Kohler–Jobin transfer, and the conclusion of Theorem 1.1.

2. Preliminaries

Throughout, $N \geq 2$, $0 < \gamma < 1$ are fixed. B_R denotes the open ball of radius R centered at the origin in $\mathbb{R}^N$, $|B_R| = \omega_N R^N$, $\omega_N = |B_1|$. All sets Ω considered are open with finite Lebesgue measure.

2.1. **Functionals.**

Torsion potential and energy. For an open bounded Ω, the torsion potential u_Ω is the unique weak solution of

$$-\Delta u_\Omega = 1 \text{ in } \Omega, \qquad u_\Omega \in H_0^1(\Omega).$$

The torsional rigidity is $T(\Omega) = \int_\Omega u_\Omega \, dx = \int_\Omega |\nabla u_\Omega|^2 \, dx$. The torsion energy is

$$E(\Omega) := \tfrac{1}{2} \int_\Omega |\nabla u_\Omega|^2 - \int_\Omega u_\Omega = -\tfrac{1}{2} T(\Omega).$$

For comparison ball B at fixed volume, $E(\Omega) \geq E(B)$ (Saint–Venant), with equality iff $\Omega = B$ (modulo null sets and translations).

Faber–Krahn data. $\lambda_1(\Omega)$ is the first Dirichlet eigenvalue of $-\Delta$ on Ω; Faber–Krahn: $\lambda_1(\Omega) \geq \lambda_1(B^*)$ at fixed volume, equality iff ball.

Asymmetry. The Fraenkel asymmetry is

$$\mathcal{A}(\Omega) := \inf_{x_0 \in \mathbb{R}^N} \frac{|\Omega \Delta (x_0 + B^*)|}{|\Omega|},$$

B^* the ball of volume $|\Omega|$ at the origin.

Quadratic asymmetry. The integrated radial asymmetry used by BDV is

$$\alpha(\Omega) := \int_{\Omega \Delta B_{r_\Omega}(x_\Omega)} \big| r_\Omega - |x - x_\Omega| \big| \, dx, \qquad r_\Omega := (|\Omega| / \omega_N)^{1/N},$$

where x_Ω is the barycenter. For unit volume this is BDV Definition 4.1. BDV Lemma 4.2 gives $\mathcal{A}(\Omega)^2 \leq C_{\mathrm{Fr}}(N) \alpha(\Omega)$.

Quotient.

$$Q_\alpha(\Omega) := \frac{E(\Omega) - E(B_1)}{\alpha(\Omega)}.$$

The whole work is about lower bounds on Q_α in the small-asymmetry regime.

2.2. **BDV regularity package.**

We use the BDV *regularity package* as a black box: it asserts that the minimizers of the penalized functionals below satisfy two-sided density, Lipschitz, nondegeneracy, and the Bernoulli coefficient bounds needed to start the flatness theory, with constants depending only on (N, R). After graph entry, the explicit selected Bernoulli law is smooth on the fixed collar and the same constants control the Schauder bootstrap. Specifically:

- **BDV Lemma 4.9** (density). There are $c_d(N, R) > 0$, $r_d(N, R) > 0$ such that for every penalized minimizer U and every free boundary point $x \in \partial U$ and $r \leq r_d$,

$$c_d r^N \leq |U \cap B_r(x)| \leq (1 - c_d) r^N \cdot \omega_N.$$

- **BDV Lemma 4.12** (Lipschitz/nondegeneracy). The torsion function u of a penalized minimizer is Lipschitz with constant $L_u(N, R)$ and nondegenerate: $\sup_{B_r(x)} u \geq c_d r$ for $x \in \partial U$, $r \leq r_d$.

- **BDV Lemma 4.16** (Bernoulli coefficient). For a penalized minimizer in the flat regime, $q_u(y) = |\nabla u(y)|$ is well-defined on ∂U and satisfies

$$q_-(N,R) \le q_u \le q_+(N,R), \qquad [q_u]_{C^{1,\gamma}(\partial U)} \le L_q(N,R).$$

- **BDV Theorem 4.18** (Alt–Caffarelli flatness improvement). If the free boundary of a penalized minimizer is $\bar\mu(N,R)$-flat at scale $\bar\rho(N,R)$, then it improves to a $C^{1,\gamma}$ graph at scale $\bar\rho/2$ with norm $C_{\mathrm{AC}}(N,R)\bar\mu$.

These four results are quoted from BDV §4. We use each one *per single set*, not along a sequence, and this is legitimate at the level of BDV's own proofs: the proofs of Lemmas 4.9–4.12 open with "being j fixed we drop the subscript" and proceed by a one-set variational comparison against the perturbed minimality inequality (4.19), which is itself the single-set first variation of Lemma 4.2; the resulting constants depend only on (N,R) and on the regularity regime $q \le q_{\mathrm{sel}}(N,R)$ (equivalently BDV's $\sigma \le \sigma_{\mathrm{reg}}(N,R)$, the penalization prefactor being $\le \tau = q^{1/4}$). The *only* place BDV pass to $j \to \infty$ is Lemma 4.13 (Kuratowski convergence $\partial\Omega_j \to \partial B_1$), invoked solely to produce a free-boundary point near every $\bar x \in \partial B_1$. That single qualitative step is exactly what Lemma 4.1 below replaces quantitatively, so Lemma 4.13 is *not invoked* anywhere in this route. The single-set quasi-minimality these per-set lemmas require is the one Lemma 3.4 secures for the selected U_* before dilation.
BDV nearly spherical theorem (used as black box). The closing step is:

> **BDV Theorem 3.3** (nearly spherical). For every $0 < \gamma_0 \le 1$ there exist $c_{\mathrm{sph}}(N) > 0$ and $\delta_{\mathrm{sph}}(N,\gamma_0) > 0$ such that for every set $U = \{(1+g(\theta))\theta : \theta \in \partial B_1\}$ with $|U| = |B_1|$, barycenter$(U) = 0$, and $\|g\|_{C^{2,\gamma_0}(\partial B_1)} \le \delta_{\mathrm{sph}}$,
>
> $$E(U) - E(B_1) \ge c_{\mathrm{sph}}(N)\, \|g\|^2_{H^{1/2}(\partial B_1)}\,.$$

This is the only step that Route A uses BDV for in the final closure; Route B would replace it by `fixed-domain-bernoulli-expansion.md`, Theorem 6.2.

3. Step 1: Selection principle and the flow heuristic

3.1. **Heuristic: the selection as a controlled deformation flow.** Before stating the single-set selection theorem itself, we explain in informal terms *why* one expects it to hold, and how its proof secretly tracks a gradient flow on a penalised energy.

Suppose we are handed a near-counterexample to Faber–Krahn, i.e. an $\Omega_0 \subset \mathbb{R}^N$ with $|\Omega_0| = |B_1|$, small deficit $\delta(\Omega_0)$, and a positive smooth asymmetry $\alpha(\Omega_0)$. Outside a regular class such an Ω_0 may be arbitrarily rough. Our goal is to replace Ω_0 by a *regular* companion $\widetilde\Omega$ whose deficit-to-asymmetry ratio $Q_\alpha(\widetilde\Omega)$ is comparable to (in fact, dominates up to a constant) $Q_\alpha(\Omega_0)$.

The BDV idea, made constructive in the present article, is to *minimise a penalised energy* of the cartoon form

$$\mathcal{J}(V) := -T(V) + \widehat\eta\left(|V| - |B_1|\right)^2 + \tau\left[\left(\alpha(V) - \alpha(\Omega_0)\right)_+\right]^2, \tag{2}$$

where $\alpha(\cdot)$ is the BDV smooth asymmetry (see the precise functional $J_{\tau,\widehat\eta,\Omega_0}$ in §3.2 below). We use α rather than $\mathcal{A}_F$ because α is Lipschitz under L^1-convergence of indicators, whereas $\mathcal{A}_F$ is only lower semicontinuous.

A minimising sequence for $\mathcal{J}$ is morally a discrete *gradient flow* V_t: along the flow the torsional term $-T(V_t)$ can only decrease, the volume penalty keeps $|V_t|$ pinned near $|B_1|$, and the asymmetry penalty $(\alpha(V)-\alpha(\Omega_0))_+^2$ only penalises drops in α below the initial value. Hence V_t cannot degenerate or escape to infinity, and the limit $\widetilde{\Omega}$ inherits the following two key estimates:

- the torsion term forces $\delta(\widetilde{\Omega}) \leq \delta(\Omega_0)$ up to the volume-penalty correction, hence $\delta(\widetilde{\Omega}) \lesssim \delta(\Omega_0)$ quantitatively;
- the asymmetry tracker, written with the one-sided positive part $(\alpha-\alpha(\Omega_0))_+$, only *penalises* drops in α, never the reverse; the minimiser therefore has $\alpha(\widetilde{\Omega}) \geq (1-\rho)\alpha(\Omega_0)$ for a controlled $\rho < 1$.

Combining the two, $Q_\alpha(\widetilde{\Omega}) \leq C_{\mathrm{sel}}/(1-\rho)\cdot Q_\alpha(\Omega_0)$, which is the quantitative replacement for BDV's sequential selection + compactness contradiction.

The limit $\widetilde{\Omega}$ is a quasi-minimiser of the *Bernoulli*-type energy $V \mapsto -T(V) + \Lambda_+|V|$ and, by the BDV/Alt–Caffarelli free-boundary package [1, Theorem 4.18], automatically has a $C^{1,\gamma}$ boundary with a class-uniform atlas norm; this regularity is the input to Steps 2 and 3 below. Geometrically the selection map acts as a *calibrated symmetrisation*: $\widetilde{\Omega}$ is the endpoint of a normal-velocity flow steered by the Bernoulli condition, replacing the asymmetric tail of Ω_0 by a regular, near-spherical piece of identical volume and at most a controlled loss of α.

The rest of this section turns the cartoon into the statement and proof of Theorem 3.2 below. The argument is a streamlined reorganisation of the construction in [1, §4], presented here so that §§4–6 may work entirely inside the structured class of selected minimisers.

3.2. **BDV's original contradiction.** BDV's proof (BDV §5) of sharp Saint–Venant runs:

1. Assume by contradiction the sharp inequality fails: there is a sequence $\{\Omega_n\}_{n\geq 1}$ in B_R with $|\Omega_n| = |B_1|$ and $Q_\alpha(\Omega_n) \to 0$.
2. For each n, apply a sequence of penalized minimizations along the sequence to extract a *selected sequence* $\{U_n\}$, with quantitative control on the energy and asymmetry losses at each step.
3. Pass to a Hausdorff limit U_∞ via the BDV regularity package (equicontinuity of free boundaries). U_∞ is a penalized minimizer with $\alpha(U_\infty) = \lim \alpha(U_n) > 0$.
4. Apply BDV Theorem 4.18 + nearly spherical to conclude $U_\infty = B_1$ (after barycenter), contradicting $\alpha(U_\infty) > 0$.

Step 3 is the contradiction step: the limit U_∞ is qualitative and the constants underlying it are not exposed.

3.3. **Single-set replacement: the selection map of `quantitative-selection-principle.md`.**

Definition 3.1 (Single-set penalized functional)**.** For $\Omega_0 \subset B_R$, $|\Omega_0| = |B_1|$, set $\varepsilon = \alpha(\Omega_0)$, $q = Q_\alpha(\Omega_0)$. For $\tau > 0$ and $\hat{\eta} > 0$, the single-set penalized functional is

$$J_{\tau,\hat{\eta},\Omega_0}(U) := E(U) + f_{\hat{\eta}}(|U|) + \sqrt{\varepsilon^2 + \tau^2(\alpha(U)-\varepsilon)^2},$$

$U \subset B_R$, where $f_{\hat{\eta}}(s)$ is the BDV symmetric volume penalization

$$f_{\hat{\eta}}(s) := \begin{cases} \hat{\eta}(s-\omega_N), & s \leq \omega_N, \\ (s-\omega_N)/\hat{\eta}, & s \geq \omega_N, \end{cases}$$

with $\hat{\eta} = \hat{\eta}(N,R)$ chosen as in BDV Lemma 4.5. This penalty does not force exact unit volume; the selected set is volume-normalized in §3.4.

Theorem 3.2 (Single-set selection, `quantitative-selection-principle.md` Thm. 1)**.** *There exist $\tau_{\mathrm{ex}}(N,R) > 0$, $\tau_{\mathrm{reg}}(N,R) > 0$, and $C_{\mathrm{sel}}(N,R) > 0$ such that for every $\tau \leq \tau_{\mathrm{reg}}(N,R)$,*

$q \le \min(q_{\mathrm{sel}}(N,R),\tau^2)$, and every $\Omega_0 \subset B_R$ with $|\Omega_0| = |B_1|$ and $Q_\alpha(\Omega_0) = q$, the functional $J_{\tau,\hat\eta,\Omega_0}$ admits a minimizer U_* with the following properties. After volume normalization (§3.4 below) to $\widetilde\Omega = r_*^{-1}U_*$, $r_* = (|U_*|\,/\,|B_1|)^{1/N}$:

(1) Existence and regularity. *U_* is a BDV penalized minimizer satisfying Lemmas 4.9, 4.12, 4.16 with constants depending only on (N,R). In particular, $\sigma_* :=$ penalization parameter of U_* satisfies $\sigma_* \le C\tau$.*

(2) Asymmetry preservation. *With $\rho(q,\tau) = \sqrt{2q+q^2}/\tau + C_{\mathrm{sel}}q$,*

$$(1-\rho(q,\tau))\,\alpha(\Omega_0) \le \alpha(\widetilde\Omega) \le (1+\rho(q,\tau))\,\alpha(\Omega_0).$$

(3) Quotient preservation.

$$Q_\alpha(\widetilde\Omega) \le \frac{C_{\mathrm{sel}}(N,R)}{1-\rho(q,\tau)}\,Q_\alpha(\Omega_0).$$

With the canonical choice $\tau = q^{1/4}$, $\rho(q,\tau) \le 1/2$ for $q \le q_{\mathrm{sel}}$.

Remark 3.3. Theorem 3.2 is the *exact single-set analogue* of BDV's sequential selection. The only change is that BDV minimize a sequence of functionals along a contradiction sequence, while here we minimize one functional given one input set. The functional uses $\varepsilon = \alpha(\Omega_0)$ as a fixed parameter, not as a sequence.

Proof sketch (full proof in `quantitative-selection-principle.md`). Existence: this is BDV Lemma 4.6 with the sequence parameter replaced by the fixed parameter τ. The proof is performed in the quasi-open class: one truncates a minimizing sequence of torsion potentials, obtains a uniform perimeter bound by the coarea argument, and then passes to the L^1/quasi-open limit. The density estimates are consequences of minimality and are not used to prove existence.

Regularity: the smooth Bernoulli term $\sqrt{\varepsilon^2+\tau^2(\alpha(U)-\varepsilon)^2}$ contributes a $C^{0,1}$ pseudonorm bounded by τ, and so falls within the BDV regularity framework with selection parameter $\sigma \le C\tau$.

Quotient preservation: by minimization, $J_{\tau,\hat\eta,\Omega_0}(U_*) \le J_{\tau,\hat\eta,\Omega_0}(\Omega_0)$. The volume penalization $f_{\hat\eta}$ forces $|U_* - B_1| \le O(\delta_T)$, $\delta_T = E(U_*) - E(B_1)$ (BDV trick). The α-bracket gives $\alpha(U_*) \ge \varepsilon - \rho\varepsilon$ with ρ as in (ii), and the quotient bound (iii) follows by combining (ii) with the energy bound $E(U_*) \le E(\Omega_0) + \text{error} \le E(B_1) + (1+O(\tau))(E(\Omega_0)-E(B_1))$. □

3.4. **Volume normalization.** The selected U_* has volume $|B_1| + O(\delta_T)$, so it must be rescaled to compare with the unit ball.

Lemma 3.4 (Volume-normalized law and quasi-minimality)**.** *Set $r_* = (|U_*|\,/\,|B_1|)^{1/N}$, $\widetilde\Omega = r_*^{-1}U_*$, $\widetilde u(x) = r_*^{-2}u_{U_*}(r_*x)$. Then*

$$|r_*-1| \le C\delta_T, \qquad \left|\widetilde\Omega\right| = |B_1|, \qquad -\Delta\widetilde u = 1 \text{ in } \widetilde\Omega,\ \widetilde u = 0 \text{ on } \partial\widetilde\Omega,$$

and the rescaled Bernoulli law on $\partial\widetilde\Omega$ reads

$$|\nabla\widetilde u(y)|^2 = \widetilde\Lambda + \widetilde a_\sigma\widetilde\Psi(y),$$

with $\widetilde\Lambda = r_*^{-2}\Lambda$, $\widetilde a_\sigma = r_*^{-1}a_\sigma$, and $\widetilde\Psi$ the rescaled α-bracket. Moreover, if U_* satisfies BDV's perturbed minimality inequality with parameter σ, then $\widetilde u$ satisfies the same type of inequality for the rescaled volume penalty

$$\widetilde f(s) := r_*^{-(N+2)} f_{\hat\eta}(r_*^N s)$$

and with quasi-minimality parameter $C\sigma r_*^{-2}$. In the regime $\delta_T \le c$ small, $r_* \in [1/2, 2]$, so all constants degrade by bounded (N, R)-factors.

Proof. The torsion equation scales by $u \mapsto r_*^{-2} u(r_* \cdot)$ and the gradient by $\nabla \widetilde{u}(y) = r_*^{-1} \nabla u_{U_*}(r_* y)$.

Write the first-variation bracket in the translation-invariant averaged form

$$A_U := \frac{1}{|U|} \int_U \frac{z - x_U}{|z - x_U|}\, dz, \qquad \Psi_U(x) := |x - x_U| - A_U \cdot (x - x_U).$$

This is BDV's bracket, up to an additive constant if one writes $A_U \cdot x$; that constant is absorbed into Λ. Under the change of variables $U_* = r_* \widetilde{\Omega}$ one has $A_{U_*} = A_{\widetilde{\Omega}}$ and

$$\Psi_{U_*}(r_* y) = r_* \Psi_{\widetilde{\Omega}}(y).$$

Thus

$$|\nabla \widetilde{u}(y)|^2 = r_*^{-2} \big(\Lambda + a_\sigma \Psi_{U_*}(r_* y)\big) = r_*^{-2} \Lambda + r_*^{-1} a_\sigma \Psi_{\widetilde{\Omega}}(y).$$

For the quasi-minimality, let ζ be an admissible competitor for $\widetilde{u}$ and set $v(x) = r_*^2 \zeta(x/r_*)$. Then

$$\mathcal{E}(v) = r_*^{N+2} \mathcal{E}(\zeta), \qquad |\{v > 0\}| = r_*^N \left|\{\zeta > 0\}\right|,$$

and

$$|U_* \Delta \{v > 0\}| = r_*^N \left|\widetilde{\Omega} \Delta \{\zeta > 0\}\right|.$$

Dividing BDV's perturbed minimality inequality for U_* by r_*^{N+2} gives the displayed rescaled inequality. ☐

Remark 3.5 (Proof order)**.** The free-boundary regularity arguments below are applied to the original selected minimizer U_* before the final dilation. Lemma 3.4 shows that applying them after dilation would also be legitimate, but the pre-dilation order avoids any hidden use of post-normalization minimality.

3.5. **Detailed selection-principle estimates.** Assume $\tau \le \tau_{\mathrm{ex}}(N, R)$, small enough that the BDV existence and perimeter estimates hold with τ in place of σ. Let U_* be a minimizer of $\mathcal{G}_{\varepsilon,\tau}$. Lemma 1. Energy and Asymmetry Preservation Before Normalization. The minimizer U_* satisfies

$$0 \le F_{\hat\eta}(U_*) - F_{\hat\eta}(B_1) \le \delta,$$

and

$$|\alpha(U_*) - \varepsilon| \le \frac{\sqrt{2\varepsilon\delta + \delta^2}}{\tau}.$$

In particular, if $q = \delta/\varepsilon \le 1$ and $\tau = q^{1/4}$, then

$$|\alpha(U_*) - \alpha(\Omega_0)| \le C \alpha(\Omega_0) q^{1/4}.$$

Proof. By minimality against Ω_0,

$$F_{\hat\eta}(U_*) + \sqrt{\varepsilon^2 + \tau^2 (\alpha(U_*) - \varepsilon)^2} \le F_{\hat\eta}(\Omega_0) + \varepsilon.$$

Since $|\Omega_0| = |B_1|$,

$$F_{\hat\eta}(\Omega_0) = E(\Omega_0) = F_{\hat\eta}(B_1) + \delta.$$

Because B_1 minimizes $F_{\hat\eta}$,

$$F_{\hat\eta}(U_*) \ge F_{\hat\eta}(B_1).$$

Therefore

$$F_{\hat\eta}(U_*) - F_{\hat\eta}(B_1) \le \delta,$$

and

$$\sqrt{\varepsilon^2 + \tau^2(\alpha(U_*) - \varepsilon)^2} \leq \varepsilon + \delta.$$

Squaring the last inequality gives

$$\tau^2(\alpha(U_*) - \varepsilon)^2 \leq 2\varepsilon\delta + \delta^2.$$

This proves the claim.

Lemma 2. Volume Error. There is $C = C(N, R)$ such that

$$\big||U_*| - |B_1|\big| \leq C\delta.$$

Proof. Let B_{r_*} be a ball with $|B_{r_*}| = |U_*|$. By rearrangement,

$$F_{\hat\eta}(B_{r_*}) \leq F_{\hat\eta}(U_*).$$

BDV Lemma 4.5 gives

$$F_{\hat\eta}(B_{r_*}) - F_{\hat\eta}(B_1) \geq \frac{|r_* - 1|}{C_4(N, R)}.$$

Together with Lemma 1,

$$|r_* - 1| \leq C\delta.$$

Since $|B_{r_*}| = \omega_N r_*^N$, this implies

$$\big||U_*| - |B_1|\big| \leq C\delta.$$

Lemma 3. Ratio Preservation After Volume Normalization. Let

$$\lambda := \left(\frac{|B_1|}{|U_*|}\right)^{1/N}, \qquad \widetilde{\Omega} := \lambda U_*.$$

After translating if needed, $\widetilde{\Omega}$ has volume $|B_1|$, and

$$\alpha(\widetilde{\Omega}) \geq \left(1 - Cq^{1/4}\right)\alpha(\Omega_0),$$

$$E(\widetilde{\Omega}) - E(B_1) \leq C\big(E(\Omega_0) - E(B_1)\big),$$

provided $q \leq q_0(N, R)$ and $\alpha(\Omega_0)$ is in the small regime. Consequently,

$$Q_\alpha(\widetilde{\Omega}) \leq CQ_\alpha(\Omega_0).$$

Proof sketch. Lemma 2 gives

$$|\lambda - 1| \leq C\delta = C\varepsilon q.$$

Since $q \ll 1$, this error is smaller than the asymmetry-preservation error $\varepsilon q^{1/4}$. The selected minimizer has the perimeter bound from the BDV existence argument, so the small dilation satisfies

$$|U_* \Delta \lambda U_*| \leq C(N, R)|\lambda - 1|.$$

The Lipschitz property of α on bounded sets then gives

$$|\alpha(\widetilde{\Omega}) - \alpha(U_*)| \leq C\varepsilon q.$$

Combining with Lemma 1 gives

$$|\alpha(\widetilde{\Omega}) - \varepsilon| \leq C\varepsilon q^{1/4}.$$

For the energy, use

$$E(\lambda U_*) = \lambda^{N+2} E(U_*),$$

the bound $F_{\hat\eta}(U_*) - F_{\hat\eta}(B_1) \leq \delta$, the Lipschitz control of $f_{\hat\eta}$ in the volume variable, and $|\lambda - 1| \leq C\delta$. Since $U_* \subset B_R$, the energy is bounded in terms of N, R, so the dilation changes the deficit by $O_{N,R}(\delta)$. Thus

$$E(\widetilde{\Omega}) - E(B_1) \le C\delta.$$

The ratio estimate follows from the two displayed bounds.

Proposition 4. Single-Set Selection Map Preserving Q_α. Fix $R \ge 2$. There are constants

$$C_{\rm sel}, P_{\rm sel}, q_{\rm sel}, \varepsilon_{\rm sel} > 0$$

depending only on N, R with the following property. Let $\Omega_0 \subset B_R$ be open or quasi-open with

$$|\Omega_0| = |B_1|, \qquad 0 < \varepsilon := \alpha(\Omega_0) \le \varepsilon_{\rm sel}, \qquad q := Q_\alpha(\Omega_0) = \frac{E(\Omega_0) - E(B_1)}{\alpha(\Omega_0)} \le q_{\rm sel}.$$

Choose

$$\tau = q^{1/4},$$

with $q_{\rm sel} \le \tau_{\rm ex}^4$, minimize $\mathcal{G}_{\varepsilon,\tau}$, and volume-normalize the minimizer by

$$\widetilde{\Omega} = \left(\frac{|B_1|}{|U_*|}\right)^{1/N} U_*.$$

Then

$$P(U_*) \le P_{\rm sel}, \qquad |\widetilde{\Omega}| = |B_1|,$$
$$\alpha(\widetilde{\Omega}) \ge \left(1 - C_{\rm sel} q^{1/4}\right) \alpha(\Omega_0) \ge \frac{1}{2}\alpha(\Omega_0),$$
$$E(\widetilde{\Omega}) - E(B_1) \le C_{\rm sel}\big(E(\Omega_0) - E(B_1)\big),$$

and therefore

$$Q_\alpha(\widetilde{\Omega}) \le 2C_{\rm sel} Q_\alpha(\Omega_0).$$

More generally, if $\tau \le \tau_{\rm ex}$ is not tied to q, set

$$\rho(q,\tau) := \frac{\sqrt{2q+q^2}}{\tau} + C_{\rm sel} q.$$

Whenever $q \le 1$, $\varepsilon \le \varepsilon_{\rm sel}$, and $\rho(q,\tau) \le 1/2$, the same argument gives

$$\alpha(\widetilde{\Omega}) \ge (1 - \rho(q,\tau))\alpha(\Omega_0), \qquad Q_\alpha(\widetilde{\Omega}) \le \frac{C_{\rm sel}}{1 - \rho(q,\tau)} Q_\alpha(\Omega_0).$$

This is the precise single-set replacement for the limiting estimate BDV obtain in Proposition 4.4(iv). The smallness of $\varepsilon_{\rm sel}$ is only used after selection, to keep the dilation inside a fixed bounded class and to make the later Hausdorff/flatness threshold available; the quotient preservation itself is governed by $\rho(q,\tau)$.

4. Step 2: Entering the boundary layer — from selected Ω to a $C^{1,\gamma}$ graph

The selected minimizer is regularized before the final dilation; equivalently, by Lemma 3.4, the volume-normalized set $\widetilde{\Omega}$ is a BDV-type quasi-minimizer with uniformly transported constants. It becomes a global graph over ∂B_1 once its asymmetry is below an explicit threshold.

4.1. **Asymmetry to Hausdorff distance.**

Lemma 4.1 (Asymmetry to Hausdorff distance)**.** *For every volume-normalized selected quasi-minimizer $\widetilde{\Omega}$ with $\left|\widetilde{\Omega}\right| = |B_1|$ and comparison ball B_1 (after translating by barycenter),*

$$d_H(\partial\widetilde{\Omega}, \partial B_1) \le C'_H(N,R)\, \alpha(\widetilde{\Omega})^{1/(2N)}.$$

Proof. Write $\alpha := \alpha(\widetilde{\Omega})$ and $d := d_H(\partial\widetilde{\Omega}, \partial B_1)$. Two per-set ingredients enter – the asymmetry–volume comparison and the two-sided density estimate – and no limit is taken anywhere.

Step 1 (asymmetry controls the symmetric difference). By BDV Lemma 4.2, for a penalized minimizer normalized by its barycenter with $\left|\widetilde{\Omega}\right| = |B_1|$,

$$\left|\widetilde{\Omega}\Delta B_1\right| \leq C_\alpha(N)\,\alpha^{1/2},$$

the exponent 1/2 accounting for the passage from the optimal Fraenkel center to the barycenter-centered ball.

Step 2 (density turns volume into one-sided distance). Fix $x \in \partial\widetilde{\Omega}$ and set $t := \operatorname{dist}(x, \partial B_1)$. Since $B_t(x)$ does not meet ∂B_1 it lies either inside B_1 or in $\mathbb{R}^N \setminus \overline{B_1}$. Put $r := \min\{t, r_d\}$ with $r_d = r_d(N,R)$ the density radius of the package. If $B_t(x) \subset B_1$, the lower density of the *complement* at the free-boundary point x (the upper-density half of BDV Lemma 4.9) gives

$$\left|B_1 \setminus \widetilde{\Omega}\right| \;\geq\; \left|\widetilde{\Omega}^c \cap B_r(x)\right| \;\geq\; c_d\,\omega_N r^N;$$

if instead $B_t(x) \subset \mathbb{R}^N \setminus \overline{B_1}$, the lower density of $\widetilde{\Omega}$ gives $\left|\widetilde{\Omega} \setminus B_1\right| \geq c_d\,\omega_N r^N$. In either case

$$\left|\widetilde{\Omega}\Delta B_1\right| \;\geq\; c_d\,\omega_N\left(\min\{t, r_d\}\right)^N.$$

Once $\alpha \leq \alpha_0(N,R)$ is small enough that $C_\alpha \alpha_0^{1/2} < c_d\,\omega_N r_d^N$, the minimum cannot be attained at r_d, so it is attained at t and, with Step 1,

$$\sup_{x\in\partial\widetilde{\Omega}} \operatorname{dist}(x, \partial B_1) \;\leq\; \left(\tfrac{C_\alpha}{c_d\,\omega_N}\right)^{1/N} \alpha^{1/(2N)}.$$

Step 3 (the reverse inclusion, no density needed). Let $\bar{x} \in \partial B_1$ and $s := \operatorname{dist}(\bar{x}, \partial\widetilde{\Omega})$. The ball $B_s(\bar{x})$ contains no point of $\partial\widetilde{\Omega}$, hence lies entirely in $\widetilde{\Omega}$ or entirely in its complement. For $s \leq \frac{1}{2}$ each of the two caps $B_s(\bar{x}) \cap B_1$ and $B_s(\bar{x}) \setminus \overline{B_1}$ has volume at least $c_N s^N$, $c_N = c_N(N) > 0$. If $B_s(\bar{x}) \subset \widetilde{\Omega}$ the outer cap lies in $\widetilde{\Omega} \setminus B_1$; if $B_s(\bar{x}) \subset \widetilde{\Omega}^c$ the inner cap lies in $B_1 \setminus \widetilde{\Omega}$. Either way $\left|\widetilde{\Omega}\Delta B_1\right| \geq c_N s^N$, so $s \leq (C_\alpha/c_N)^{1/N}\alpha^{1/(2N)}$.

Combining Steps 2–3,

$$d_H(\partial\widetilde{\Omega}, \partial B_1) \leq C'_H(N,R)\,\alpha^{1/(2N)}, \qquad C'_H = \max\Big\{\big(\tfrac{C_\alpha}{c_d\omega_N}\big)^{1/N}, \big(\tfrac{C_\alpha}{c_N}\big)^{1/N}\Big\},$$

for every $\alpha \leq \alpha_0(N,R)$, where α_0 is the minimum of the finitely many displayed smallness conditions ($C_\alpha \alpha_0^{1/2} < c_d\omega_N r_d^N$ for Step 2, and $(C_\alpha/c_N)^{1/N}\alpha_0^{1/(2N)} \leq \frac{1}{2}$ so that $s \leq \frac{1}{2}$ in Step 3). The threshold α_0 is thus a fixed (N,R)-quantity (it is absorbed into α_{graph} in §4); it depends on neither δ_T nor the conclusion. The constants c_d, r_d are the per-set density constants of the regularity package, valid for the single selected set by the discussion following its statement; no sequence or compactness is used. □

4.2. **Hausdorff to flatness.** Pick BDV Theorem 4.18 thresholds $\bar{\mu}(N,R), \bar{\rho}(N,R)$. Define $\mu(N,R) \leq \bar{\mu}/16C_0$ small enough for the rest of the chain (§5), where C_0 is the universal conversion constant between the Alt–Caffarelli slab and the $F(\cdot, 1, +\infty)$ normalization. Pick $\rho_\mu \leq \bar{\rho}$ so that every cap of ∂B_1 of radius $6\rho_\mu$ lies in a $\mu\rho_\mu$-slab around its tangent. Set $h_F := \mu\rho_\mu/16$.

Proposition 4.2 (Hausdorff closeness to Alt–Caffarelli flatness)**.** *If $d_H(\partial\widetilde{\Omega}, \partial B_1) \leq h_F$, then for every $\bar{x} \in \partial B_1$ there is $x_0 \in \partial\widetilde{\Omega} \cap B_{h_F}(\bar{x})$ such that in $B_{4\rho_\mu}(x_0)$, the torsion function $\widetilde{u}$ is of Alt–Caffarelli class $F(C_0\mu, 1, +\infty)$ with respect to the tangent direction $\nu_{\bar{x}}$.*

Proof. First, Hausdorff closeness and $\left|\widetilde{\Omega}\right| = |B_1|$ give the annular sandwich

$$B_{1-h_F} \subset \widetilde{\Omega} \subset B_{1+h_F}.$$

The outer inclusion follows because a point of $\widetilde{\Omega}$ outside B_{1+h_F} would force a boundary point farther than h_F from ∂B_1. For the inner inclusion, any component of $B_{1-h_F} \setminus \widetilde{\Omega}$ would have boundary inside B_{1-h_F}, again contradicting the Hausdorff bound; the alternative $B_{1-h_F} \subset \mathbb{R}^N \setminus \widetilde{\Omega}$ contradicts the volume identity for h_F small, since $\widetilde{\Omega} \subset B_{1+h_F}$.

Choose $x_0 \in \partial\widetilde{\Omega} \cap B_{h_F}(\bar{x})$ by the Hausdorff bound. The curvature of ∂B_1 gives

$$\partial B_1 \cap B_{5\rho_\mu}(\bar{x}) \subset \{x : |(x-\bar{x})\cdot \nu_{\bar{x}}| \le C\rho_\mu^2\}.$$

Combining this with the Hausdorff bound and recentering at x_0 yields

$$\partial\widetilde{\Omega} \cap B_{4\rho_\mu}(x_0) \subset \{x : |(x-x_0)\cdot \nu_{\bar{x}}| \le C(\rho_\mu^2 + h_F)\}.$$

The choices $\rho_\mu \le c\mu$ and $h_F \le c\mu\rho_\mu$ make the last height at most $C\mu\rho_\mu$.

With $\nu_{\bar{x}}$ oriented as the inward normal to B_1, the slab bound and the annular sandwich pin *both* phases deterministically. A point $x \in B_{4\rho_\mu}(x_0)$ with $(x-x_0)\cdot\nu_{\bar{x}} \le -C_0\mu(4\rho_\mu)$ lies outside B_{1+h_F} once C_0 is chosen large, hence, by the outer inclusion $\widetilde{\Omega} \subset B_{1+h_F}$ of the sandwich, satisfies $\widetilde{u}(x) = 0$. Symmetrically, a point with $(x-x_0)\cdot\nu_{\bar{x}} \ge +C_0\mu(4\rho_\mu)$ lies inside B_{1-h_F}, hence, by the inner inclusion $B_{1-h_F} \subset \widetilde{\Omega}$, lies in $\widetilde{\Omega}$, so $\widetilde{u} > 0$ there. Thus the positive phase of the class $F(C_0\mu, 1, +\infty)$ is *not* an additional hypothesis to be assumed: it is supplied, with the same explicit constants, by the inner inclusion of the sandwich, exactly mirroring the empty outer half-slab.

The only quantitative input the flatness-improvement iteration consumes on the positive side is a growth rate, and this is the *explicit per-set* nondegeneracy of the regularity package: for every $x_0 \in \partial\widetilde{\Omega}$ and $r \le r_d(N, R)$,

$$\sup_{B_r(x_0)} \widetilde{u} \;\ge\; c_d(N,R)\, r,$$

(BDV Lemma 4.12; the constant is $c_d(N, R)$, never relative to δ_T), together with the two-sided density $c_d\,\omega_N r^N \le \left|\widetilde{\Omega} \cap B_r(x_0)\right| \le (1-c_d)\,\omega_N r^N$ of BDV Lemma 4.9. Both hold for the single selected set under $q \le q_{\mathrm{sel}}(N, R)$, by the per-set discussion in §2. With the zero set, the positive set, and the nondegeneracy/density constants all exhibited, $\widetilde{u}$ is of Alt–Caffarelli class $F(C_0\mu, 1, +\infty)$ in $B_{4\rho_\mu}(x_0)$ with the package's (N, R)-constants. ☐

4.3. **Local flatness to global graph.**

Proposition 4.3 (Local flatness to a global radial graph)**.** *Assume Prop. 4.2 holds at every $\bar{x} \in \partial B_1$. Choose μ small enough that $C_0\mu \le \bar{\mu}$ (so BDV 4.18 applies) and $C_{\mathrm{AC}}\mu \le \mu_{\mathrm{tub}}(N)$, the tubular-neighborhood slope for radial projection. Then*

$$\partial\widetilde{\Omega} = \{(1+g(\theta))\theta : \theta \in \partial B_1\}$$

with

$$\|g\|_{L^\infty(\partial B_1)} \le C(N,R)\, d_H(\partial\widetilde{\Omega}, \partial B_1), \quad \|g\|_{C^{1,\gamma}(\partial B_1)} \le C(N,R)\mu.$$

Proof. BDV Theorem 4.18 in each $B_{\rho_\mu}(x_0)$ of Prop. 4.2 gives local $C^{1,\gamma}$ graphs of norm $\le C_{\mathrm{AC}}\mu$. These charts cover the entire free boundary: if $y \in \partial\widetilde{\Omega}$, choose $\bar{x} = y/|y| \in \partial B_1$; the annular sandwich gives $|y - \bar{x}| \le h_F$, while the corresponding chart center x_0 satisfies $|x_0 - \bar{x}| \le h_F$, so $y \in B_{\rho_\mu}(x_0)$ after shrinking h_F/ρ_μ.

On every local chart the radial projection is a $C^{1,\gamma}$ diffeomorphism onto its image, because the local normal makes an angle bounded away from $\pi/2$ with the radial direction when $C_{\mathrm{AC}}\mu \leq \mu_{\mathrm{tub}}$. Surjectivity of the global radial projection follows from the annular sandwich: every ray from the origin meets $\partial\widetilde{\Omega}$ between radii $1-h_F$ and $1+h_F$. For injectivity, if two boundary points lay on the same ray, their distance would be at most $2h_F < \rho_\mu/2$, so both would lie in one local chart; a transverse local graph is intersected by such a radial line at most once. Hence the intersection on every ray is unique.

The unique radial intersection defines g. The L^∞ bound follows from the annular sandwich, and the $C^{1,\gamma}$ bound follows by passing the finite family of local chart estimates through the uniformly nondegenerate radial-projection coordinates. □

4.4. **Graph-entry threshold.**

Theorem 4.4 (Graph-entry threshold)**.** *Set*

$$\alpha_{\mathrm{graph}}(N,R) := \Big(\frac{h_F}{C'_H(N,R)}\Big)^{2N}.$$

Every volume-normalized selected quasi-minimizer $\widetilde{\Omega} \subset B_R$ with $\big|\widetilde{\Omega}\big| = |B_1|$, barycenter 0, and $\alpha(\widetilde{\Omega}) \leq \alpha_{\mathrm{graph}}$ is a $C^{1,\gamma}$ normal graph over ∂B_1 with $\|g\|_{C^{1,\gamma}} \leq C(N,R)\mu(N,R)$.

Proof. Combine Lemmas 4.1, Propositions 4.2, 4.3. □

5. Step 3: Realising the inset as a small C^{2,γ_0} graph

BDV Theorem 3.3 needs the graph norm to be small in C^{2,γ_0} for some $0 < \gamma_0 \leq 1$. Graph entry gives small $C^{1,\gamma}$ norm and, more importantly, small L^∞ norm through Hausdorff closeness. The Schauder bootstrap supplies uniform higher norms; interpolation then turns small L^∞ plus uniform high regularity into small C^{2,γ_0}.

Proposition 5.1 (Uniform bootstrap)**.** *Let $\partial\widetilde{\Omega} = \{(1+g)\theta\}$ be a graph obtained from Theorem 4.4, and let $\widetilde{u}$ be its torsion potential. For every integer $m \geq 3$ there is $M_m(N,R) < \infty$ such that*

$$\|g\|_{C^{m,\gamma}(\partial B_1)} \leq M_m(N,R).$$

Proof. Work first with the unscaled selected minimizer U_*, and write $|U_*| = |B_r|$, with $r \in [1/2, 2]$ in the small-deficit regime. The scale-invariant form of the graph-entry step gives a $C^{1,\gamma}$ graph over $\partial B_r(x_{U_*})$ and the annular containment

$$B_{r-h}(x_{U_*}) \subset U_* \subset B_{r+h}(x_{U_*}), \qquad h \leq r/4.$$

Hence a fixed collar of the free boundary lies in $\{r/2 \leq |x - x_{U_*}| \leq 3r/2\}$.

The selected Bernoulli coefficient has the form

$$q_{U_*}(x)^2 = \Lambda + a_\sigma \Psi_{U_*}(x), \qquad \Psi_{U_*}(x) = |x - x_{U_*}| - A_{U_*} \cdot (x - x_{U_*}),$$

with $|A_{U_*}| \leq 1$, $|a_\sigma| \leq C\sigma$, and $0 < q_- \leq q_{U_*} \leq q_+$. On the fixed annular collar,

$$\|\Psi_{U_*}\|_{C^k} \leq C_k(N,R)$$

for every fixed k, since only derivatives of $|x - x_{U_*}|$ and a linear term occur. Choosing σ below BDV's regularity threshold and using the positive lower bound on q_{U_*} gives

$$\|q_{U_*}\|_{C^k(\mathrm{collar})} \leq C_k(N,R).$$

This is the single-set content of BDV Lemma 4.16; no derivative of the unknown graph enters this estimate.

Now apply the higher-regularity clause of BDV Theorem 4.18. The local Alt–Caffarelli charts obtained in Theorem 4.4 have $C^{m,\gamma}$ norms bounded by a constant depending only on N, R, m and the above $C^{m-1,\gamma}$ bound for q_{U_*}. A finite cover of $\partial B_r(x_{U_*})$, together with the already-known small C^1 slope, turns these local estimates into a global radial graph bound

$$\|g_r\|_{C^{m,\gamma}(\partial B_r)} \le M'_m(N, R).$$

The final dilation in Lemma 3.4 changes this norm by at most a bounded factor because $r \in [1/2, 2]$. This gives the claimed $M_m(N, R)$ for $\widetilde{\Omega}$. □

Lemma 5.2 (Interpolation to the nearly spherical threshold)**.** *Fix* $0 < \gamma_0 < \gamma$ *and choose* $m \ge 3$*. There are* $h_{\mathrm{sph}}(N, R, \gamma_0) > 0$ *and*

$$\alpha_{\mathrm{sph}}(N, R, \gamma_0) := \left(\frac{h_{\mathrm{sph}}}{C_\infty C'_H(N, R)} \right)^{2N}$$

such that every selected graph with $\alpha(\widetilde{\Omega}) \le \alpha_{\mathrm{sph}}$ *satisfies*

$$\|g\|_{C^{2,\gamma_0}(\partial B_1)} \le \delta_{\mathrm{sph}}(N, \gamma_0),$$

the smallness hypothesis of BDV Theorem 3.3.

Proof. Graph entry gives

$$\|g\|_{L^\infty(\partial B_1)} \le C_\infty d_H(\partial\widetilde{\Omega}, \partial B_1) \le C_\infty C'_H(N, R)\alpha(\widetilde{\Omega})^{1/(2N)}.$$

The interpolation inequality on ∂B_1 gives, for some $\theta = \theta(N, m, \gamma_0) \in (0, 1)$,

$$\|g\|_{C^{2,\gamma_0}} \le C\ \|g\|_{L^\infty}^{1-\theta}\ \|g\|_{C^{m,\gamma}}^{\theta} \le C\ \|g\|_{L^\infty}^{1-\theta}\ M_m(N, R)^\theta.$$

Choose h_{sph} so that the last quantity is at most $\delta_{\mathrm{sph}}(N, \gamma_0)$ whenever $\|g\|_{L^\infty} \le h_{\mathrm{sph}}$. □

5.1. **Bernoulli spectral closure: an alternative completion of Step 3 (conditional).** This subsection records an alternative pathway from the $C^{1,\gamma}$ graph output of Step 2 to the quantitative closure of Step 4. The small-C^{2,γ_0} bootstrap proved above feeds the BDV nearly-spherical second-variation theorem [1, Theorem 3.3]; the Bernoulli spectral closure instead exploits the Bernoulli boundary law $|\nabla u_{\widetilde{\Omega}}| = \Lambda$ inherited by the selected $\widetilde{\Omega}$ to derive a H^1 coercivity inequality for the normal-displacement function $g : \mathbb{S}^{N-1} \to \mathbb{R}$ that parametrises $\partial\widetilde{\Omega}$ over ∂B_1.

The pathway is genuinely *alternative* (it does not factor through BDV Theorem 3.3) and *conditional*: it requires the boundedness of a specific bilinear remainder $S_1 + S_2$ arising in the fixed-domain IFT expansion of the torsion equation; this enumeration is partial (see [7, §3] for the current status). The route is the BDV-style analog of the "hole-filling" polar-slicing alternative recorded in the companion Selection–Level–Brandolini manuscript: a slick, geometrically transparent closure that works in a restricted setting and avoids the small-C^{2,γ_0} bootstrap entirely.

Setup of the Bernoulli closure. After the selection of Step 1 and the volume normalisation, $\widetilde{\Omega}$ admits a Bernoulli boundary law: $|\nabla u_{\widetilde{\Omega}}| = \Lambda$ on $\partial\widetilde{\Omega}$ for an explicit constant $\Lambda = \Lambda(N, R) > 0$. After the graph entry of Step 2, $\partial\widetilde{\Omega}$ is parametrised by a $C^{1,\gamma}$ function $g : \mathbb{S}^{N-1} \to \mathbb{R}$ via the normal-graph map $\partial\widetilde{\Omega} = \{(1 + g(\omega))\omega : \omega \in \mathbb{S}^{N-1}\}$.

Define the *pulled-back torsion* $\widetilde{u}(g) : B_1 \to \mathbb{R}$ as the solution of $-\Delta\widetilde{u} = 1$ in B_1 with boundary condition prescribed by the change-of-variables of the original torsion equation under the diffeomorphism $\Phi_g : B_1 \to \widetilde{\Omega}$, $\Phi_g(r\omega) := (1 + rg(\omega))\, r\omega$. The first variation of $\widetilde{u}$ in g at $g = 0$ is

$$\widetilde{u}'(g)(x) \;=\; u_1(x) - \chi(r)\, \frac{r\, g(\theta)}{N}, \tag{3}$$

where u_1 is an explicit harmonic correction and χ a smooth radial cutoff; this is the fixed-domain analog of the BDV exterior-harmonic expansion, but constructed entirely on $\overline{B_1}$ with no exterior continuation. The implicit function theorem applied to the Bernoulli boundary equation $|\nabla \widetilde{u}(g)| = \Lambda$ produces a $C^{2,\gamma}$ quadratic Schauder remainder $R_2(g)$ and a bilinear remainder $S_1(g, h) + S_2(g, h)$ acting on $C^{2,\gamma} \times H^1 \to L^2$.

The spectral lemma. On the orthogonal complement of $\operatorname{span}\{1, \theta_1, \ldots, \theta_N\}$ (volume and barycentre modes, killed by the normalisations of Step 1), the linearised Bernoulli operator $\mathcal{L} := k - 1$ acting on degree-k spherical harmonics gains a full derivative:

$$\|\mathcal{L}h\|_{L^2(\mathbb{S}^{N-1})} \;\gtrsim\; \|h\|_{H^1(\mathbb{S}^{N-1})}, \qquad h \in P_{\geq 2} H^1(\mathbb{S}^{N-1}), \tag{4}$$

where $P_{\geq 2}$ is the projection onto modes of degree ≥ 2.

The conditional closure.

Theorem 5.3 (Bernoulli spectral closure, conditional)**.** *Assume the bilinear source enumeration of S_1, S_2 produces the remainder bound $\|S_i(g, h)\|_{L^2(B_1)} \leq C_S(N, R)\, \|g\|_{C^{2,\gamma}}\, \|h\|_{H^1}$ with an explicit $C_S(N, R)$. Then under the hypotheses of Theorem 5.1 below (small $\|g\|_{C^{2,\gamma}}$ and barycentre normalisation),*

$$\delta_T(\widetilde{\Omega}) \;\geq\; c_{\mathrm{Br}}(N, R)\, \|g\|^2_{H^1(\mathbb{S}^{N-1})},$$

with $c_{\mathrm{Br}}(N, R) > 0$ explicit in terms of the Steklov constant (4), the linear-correction constant in (3), and the bilinear constant $C_S(N, R)$. Consequently $\mathcal{A}_F(\widetilde{\Omega})^2 \leq C'_{\mathrm{Br}}(N, R)\, \delta_T(\widetilde{\Omega})$, and the rest of the chain (transfer back to Ω_0 via Theorem 3.2, Kohler–Jobin transfer of §6.4) concludes Theorem 1.1 without invoking BDV Theorem 6.1.

A full proof and the enumeration of S_1, S_2 are in the companion note `fixed-domain-bernoulli-expansion.t` of the working Plan 1 archive; we refer the reader there for the explicit constants.

Status remark. The unconditional closure of §§6–6.1 above uses BDV Theorem 3.3 as a black box; it requires only the small-C^{2,γ_0} output of Proposition 5.1. The Bernoulli spectral closure of this subsection is *independent* of BDV Theorem 3.3 and operates only at the $C^{2,\gamma} \times H^1$ level, which is more natural from the Bernoulli viewpoint and quantitatively explicit in (N, R). Both routes yield $\mathcal{A}_F^2 \lesssim \delta_T$ with the sharp exponent 2. The unconditional one is the route used to assemble Theorem 1.1.

6. Step 4: Quantitative closure for L^∞–L^2 small graphs

Theorem 6.1 (BDV nearly spherical, restated as black box)**.** ***(BDV Theorem 3.3.)*** *For every $0 < \gamma_0 \leq 1$ there exist $c_{\mathrm{sph}}(N) > 0$ and $\delta_{\mathrm{sph}}(N, \gamma_0) > 0$ such that for every set U of the form $\partial U = \{(1 + g(\theta))\theta : \theta \in \partial B_1\}$ with*

- $|U| = |B_1|$ *(volume normalization),*
- $\int_{\partial B_1} g\, \theta\, d\theta = 0$ *(barycenter normalization),*
- $\|g\|_{C^{2,\gamma_0}(\partial B_1)} \leq \delta_{\mathrm{sph}}(N, \gamma_0)$,

we have

$$E(U) - E(B_1) \geq c_{\mathrm{sph}}(N)\, \|g\|^2_{H^{1/2}(\partial B_1)}.$$

This is BDV Theorem 3.3 verbatim. We use it without modification.

Lemma 6.2 (Asymmetry vs. L^2 graph norm, BDV Lemma 4.2 + 5.2). *For sets U of the form above with $|U| = |B_1|$ and $\|g\|_{C^0(\partial B_1)} \le \delta_0$ small,*

$$\alpha(U) \le C_{\mathrm{BDV}}(N)\, \|g\|^2_{L^2(\partial B_1)}.$$

Proof. Direct computation: $\alpha(U) = \int_U (|x| - 1)\, dx = \int_{\partial B_1} \int_0^{1+g} (r-1) r^{N-1}\, dr\, d\theta = \int_{\partial B_1} \big[\frac{1}{2} g^2 + O(g^3)\big]\, d\theta$. The $O(g^3)$ tail is dominated by $\|g\|_{C^0} \|g\|^2_{L^2} \le \delta_0 \|g\|^2_{L^2}$. □

Corollary 6.3 (Quotient lower bound for graph minimizers). *For every U as in Theorem 6.1,*

$$Q_\alpha(U) = \frac{E(U) - E(B_1)}{\alpha(U)} \ge c_*(N) := \frac{c_{\mathrm{sph}}(N)}{C_{\mathrm{BDV}}(N)}.$$

Proof. Combine Theorem 6.1 and Lemma 6.2, noting $\|g\|^2_{L^2} \le \|g\|^2_{H^{1/2}}$ (interpolation on ∂B_1). □

6.1. **The non-contradictory quantitative argument.** We now assemble Theorem 3.2, Theorem 4.4, Proposition 5.1, Lemma 5.2, and Corollary 6.3 into a direct lower bound on $Q_\alpha(\Omega_0)$.

Set

$$\alpha_{\mathrm{small}}(N, R) := \frac{1}{2} \min\{\varepsilon_{\mathrm{sel}}, \alpha_{\mathrm{graph}}, \alpha_{\mathrm{sph}}\},$$

where $\varepsilon_{\mathrm{sel}}$ is the smallness threshold in the selection lemma and α_{sph} is Lemma 5.2's interpolation threshold.

Theorem 6.4 (Sharp Saint–Venant stability, single-set form). *There exists $q_*(N, R) > 0$ such that for every open $\Omega_0 \subset B_R$ with $|\Omega_0| = |B_1|$, $\alpha(\Omega_0) \le \alpha_{\mathrm{small}}(N, R)$, and $Q_\alpha(\Omega_0) < q_{\mathrm{sel}}(N, R)$,*

$$Q_\alpha(\Omega_0) \ge q_*(N, R) = \frac{c_*(N)}{2C_{\mathrm{sel}}(N, R)},$$

or equivalently

$$E(\Omega_0) - E(B_1) \ge q_*(N, R)\, \alpha(\Omega_0).$$

Proof. Fix Ω_0 as in the hypotheses, $q_0 := Q_\alpha(\Omega_0) < q_{\mathrm{sel}}$. Apply Theorem 3.2 with $\tau = q_0^{1/4}$; then $\tau \le q_{\mathrm{sel}}^{1/4} \le \tau_{\mathrm{reg}}$ for q_{sel} small enough, and $\rho(q_0, \tau) \le 1/2$. We obtain a volume-normalized selected set $\widetilde{\Omega} \subset B_R$ (after Lemma 3.4) with

$$\alpha(\widetilde{\Omega}) \ge \tfrac{1}{2}\alpha(\Omega_0), \quad \alpha(\widetilde{\Omega}) \le \tfrac{3}{2}\alpha(\Omega_0), \quad Q_\alpha(\widetilde{\Omega}) \le 2C_{\mathrm{sel}}(N, R)\, q_0, \tag{5}$$

selection parameter $\sigma \le C q_0^{1/4}$.

Graph entry. Since $\alpha(\Omega_0) \le \alpha_{\mathrm{small}}$, (5) gives

$$\alpha(\widetilde{\Omega}) \le \frac{3}{2}\alpha_{\mathrm{small}} \le \alpha_{\mathrm{graph}}, \qquad \alpha(\widetilde{\Omega}) \le \alpha_{\mathrm{sph}}.$$

By Theorem 4.4, $\partial\widetilde{\Omega} = \{(1+g)\theta\}$ is a $C^{1,\gamma}$ graph with $\|g\|_{C^{1,\gamma}} \le C\mu$. By Lemma 5.2, $\|g\|_{C^{2,\gamma_0}} \le \delta_{\mathrm{sph}}(N, \gamma_0)$.

Nearly spherical and quotient bound. By Corollary 6.3, $Q_\alpha(\widetilde{\Omega}) \ge c_*(N)$.

Combining. (5) gives $Q_\alpha(\widetilde{\Omega}) \le 2C_{\mathrm{sel}} q_0$, while the previous step gives $Q_\alpha(\widetilde{\Omega}) \ge c_*(N)$. Hence

$$q_0 \ge \frac{c_*(N)}{2C_{\mathrm{sel}}(N, R)} =: q_*(N, R).$$

□

Remark 6.5 (Why this is not a contradiction). At no point did we assume $Q_\alpha(\Omega_0)$ is arbitrarily small. We *computed* the inequality $Q_\alpha(\Omega_0) \ge c_*(N)/(2C_{\mathrm{sel}}(N, R))$ directly from the chain of constants. The selection is single-set; the graph entry is below an explicit threshold $\alpha_{\mathrm{graph}}(N,R)$; the closure is the pointwise BDV nearly spherical inequality.

The compactness/limit step of BDV is replaced by Theorem 3.2, which produces $\widetilde{\Omega}$ from a single Ω_0 with explicit (N,R)-constants.

6.2. Extension to all asymmetries.

Theorem 6.6 (Sharp Saint–Venant, global form)**.** *There exists $c_{\mathrm{SV}}(N,R)>0$ such that for every open $\Omega\subset B_R$ with $|\Omega|=|B_1|$,*

$$E(\Omega)-E(B_1)\geq c_{\mathrm{SV}}(N,R)\,\mathcal{A}(\Omega)^2.$$

Explicitly,

$$c_{\mathrm{SV}}(N,R)=\min\Big(\frac{q_*(N,R)}{C_{\mathrm{Fr}}(N)},\ c_{\mathrm{far,SV}}(N,R)\Big),$$

where $C_{\mathrm{Fr}}(N)$ is BDV Lemma 4.2's constant and $c_{\mathrm{far,SV}}(N,R)$ is the explicit far-from-ball lower bound from §6.3 below.

Proof. Small asymmetry, $\alpha(\Omega)\leq\alpha_{\mathrm{small}}$: combine Theorem 6.4 with $\mathcal{A}^2\leq C_{\mathrm{Fr}}\alpha$.

Large asymmetry, $\alpha(\Omega)>\alpha_{\mathrm{small}}$: direct compactness, see §6.3.

Take $c_{\mathrm{SV}}:=$ min of the two constants. □

6.3. The far-from-ball Saint–Venant constant.

Lemma 6.7 (Far-from-ball uniform lower bound)**.** *There exists $c_{\mathrm{far,SV}}(N,R)>0$ such that for every open $\Omega\subset B_R$ with $|\Omega|=|B_1|$ and $\alpha(\Omega)\geq\alpha_{\mathrm{small}}(N,R)$,*

$$E(\Omega)-E(B_1)\geq c_{\mathrm{far,SV}}(N,R).$$

Proof. First convert α-separation into Fraenkel separation. Let B be an optimal ball for $\mathcal{A}(\Omega)$. Since $\Omega\subset B_R$ and $|B|=|B_1|$, the relevant balls are contained in a fixed ball B_{R+2} after enlarging R by a dimensional amount. BDV Lemma 4.2(ii), applied to Ω and B, gives

$$\alpha(\Omega)\leq C_2(N,R)\,|\Omega\Delta B|=C_2(N,R)|B_1|\mathcal{A}(\Omega).$$

Thus $\alpha(\Omega)\geq\alpha_{\mathrm{small}}$ implies

$$\mathcal{A}(\Omega)\geq\alpha_{\mathrm{small}}/(C_2(N,R)|B_1|)=:a_{\mathrm{far}}(N,R)>0.$$

Now use the suboptimal quantitative Saint–Venant inequality quoted by BDV as Lemma 5.2:

$$E(\Omega)-E(B_1)\geq C_8(N)^{-1}\mathcal{A}(\Omega)^4.$$

Therefore one can take

$$c_{\mathrm{far,SV}}(N,R):=C_8(N)^{-1}a_{\mathrm{far}}(N,R)^4.$$

□

Remark 6.8. The Lemma 6.7 infimum depends on (N,R) but is not given explicitly. For numerical applications, $c_{\mathrm{far,SV}}$ can be estimated by known sharp inequalities for specific classes (convex bodies, smooth domains), but for the theorem above we only need positivity.

6.4. Kohler–Jobin transfer to Faber–Krahn.

Theorem 6.9 (Kohler–Jobin inequality, 1978)**.** *For every open bounded $\Omega\subset\mathbb{R}^N$ with $\lambda_1(\Omega)<\infty$,*

$$\Psi(\Omega):=T(\Omega)\cdot\lambda_1(\Omega)^{(N+2)/2}\geq\Psi(B),$$

where B is any ball, with equality iff Ω is a ball.

Lemma 6.10 (Pointwise transfer at fixed volume)**.** *For every open Ω with $|\Omega| = |B^*|$,*

$$\lambda_1(\Omega) \geq \lambda_1(B^*)\Big(\frac{T(B^*)}{T(\Omega)}\Big)^{2/(N+2)}.$$

Lemma 6.11 (Small-deficit linearization)**.** *Let $\delta_{\mathrm{SV}} := T(B^*) - T(\Omega)$. For $\delta_{\mathrm{SV}} \leq T(B^*)/2$,*

$$\lambda_1(\Omega) - \lambda_1(B^*) \geq \frac{2\lambda_1(B^*)}{(N+2)T(B^*)}\,\delta_{\mathrm{SV}}.$$

Proof. For $t \in [0, 1/2]$, $(1-t)^{-1/\beta} - 1 \geq t/\beta$ where $\beta = (N+2)/2$. Apply with $t = \delta_{\mathrm{SV}}/T(B^*)$ and use $\lambda_1(\Omega) \geq L \cdot (1-t)^{-1/\beta}$. □

Remark 6.12 (The factor $2L_N/((N+2)T_N)$)**.** $L_N := \lambda_1(B_1) = j^2_{N/2-1,1}$ where $j_{\nu,1}$ is the first positive zero of the Bessel function J_ν. $T_N := T(B_1) = |B_1|/(N(N+2))$. For $N = 2$: $L_2 = j^2_{0,1} \approx 5.78$, $T_2 = \pi/8$, multiplier $\approx 2 \cdot 5.78/(4 \cdot \pi/8) \approx 3.68/\pi$. For $N = 3$: $L_3 = \pi^2$, $T_3 = 4\pi/15$, multiplier $\approx 2\pi^2/(5 \cdot 4\pi/15) = 3\pi/2$. These are universal in N, with no R-dependence.

6.5. The sharp Faber–Krahn theorem.

Theorem 6.13 (Sharp Faber–Krahn stability, Route A end-to-end)**.** *There exists $c_{\mathrm{FK}}(N, R) > 0$ such that for every open $\Omega \subset B_R$ with $|\Omega| = |B^*|$,*

$$\lambda_1(\Omega) - \lambda_1(B^*) \geq c_{\mathrm{FK}}(N, R)\,\mathcal{A}(\Omega)^2.$$

Proof. Small asymmetry. By Theorem 6.6, in the small-asymmetry regime $\mathcal{A}(\Omega) \leq \mathcal{A}_*(N, R)$, $\delta_{\mathrm{SV}} := T(B^*) - T(\Omega) = 2(E(\Omega) - E(B^*)) \geq 2c_{\mathrm{SV}}(N, R)\mathcal{A}(\Omega)^2$ (where $E = -T/2$). By Lemma 6.11, for δ_{SV} small,

$$\lambda_1(\Omega) - \lambda_1(B^*) \geq \frac{2L_N}{(N+2)T_N}\,\delta_{\mathrm{SV}} \geq \frac{4L_N\, c_{\mathrm{SV}}(N, R)}{(N+2)T_N}\,\mathcal{A}(\Omega)^2.$$

Large asymmetry or large torsion deficit. If $\mathcal{A}(\Omega) > \mathcal{A}_*$, the suboptimal Faber–Krahn stability quoted by BDV gives a positive lower bound depending only on (N, R); call it $c_{\mathrm{far,FK}}(N, R)$. If instead $\delta_{\mathrm{SV}} > T_N/2$, the Kohler–Jobin pointwise bound gives

$$\lambda_1(\Omega) - L_N \geq L_N(2^{2/(N+2)} - 1),$$

which is also bounded below by a dimensional multiple of $\mathcal{A}(\Omega)^2$ because $\mathcal{A} \leq 2$.

Take

$$c_{\mathrm{FK}}(N, R) := \min\Big(\frac{4L_N\, c_{\mathrm{SV}}(N, R)}{(N+2)T_N},\ \frac{c_{\mathrm{far,FK}}(N, R)}{4},\ \frac{L_N(2^{2/(N+2)} - 1)}{4}\Big).$$

□

Remark 6.14 (Full constant chain)**.** The constants are produced in the order:

1. Selection: $C_{\mathrm{sel}}(N, R), q_{\mathrm{sel}}(N, R), \tau_{\mathrm{reg}}(N, R)$ from `quantitative-selection-principle.md` (depending on BDV Lemmas 4.9, 4.12, 4.16).
2. Graph entry: $\alpha_{\mathrm{graph}}(N, R)$ from $\alpha_{\mathrm{graph}} = (h_F/C'_H)^{2N}$, where $h_F = \mu\rho_\mu/16$ and μ, ρ_μ are BDV Theorem 4.18 thresholds.
3. Bootstrap: $C_{\mathrm{boot}}(N, R)$ from Kinderlehrer–Nirenberg / Lieberman applied with BDV Lemma 4.16's L_q.
4. Nearly spherical: $c_{\mathrm{sph}}(N), \delta_{\mathrm{sph}}(N, \gamma_0)$ from BDV Theorem 3.3.

(5) Asymmetry vs. L^2: $C_{\mathrm{BDV}}(N)$ from BDV Lemma 4.2 + 5.2.
(6) Saint–Venant quotient: $c_*(N) = c_{\mathrm{sph}}(N)/C_{\mathrm{BDV}}(N)$.
(7) Saint–Venant constant: $q_*(N,R) = c_*(N)/(2C_{\mathrm{sel}}(N,R))$.
(8) Global Saint–Venant: $c_{\mathrm{SV}}(N,R) = \min(q_*/C_{\mathrm{Fr}}, c_{\mathrm{far,SV}})$.
(9) Kohler–Jobin multiplier: $2L_N/((N+2)T_N)$, universal.
(10) Faber–Krahn: $c_{\mathrm{FK}}(N,R)$ is the minimum displayed in Theorem 6.13.

Every constant in this chain depends only on (N,R), and the dependence is through named BDV regularity constants. No constant arises from a limit or compactness argument that hides its (N,R)-dependence; the only far-regime constants are $c_{\mathrm{far,SV}}$ and $c_{\mathrm{far,FK}}$, which are obtained from suboptimal stability bounds away from the ball and do not enter the small-asymmetry regime.

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